

Frank Goortani

Staff AI/ML Engineer • Distributed Systems • Production LLM Systems

Aurora, Ontario, Canada (Remote) • frank.goortani@gmail.com • (647) 983-5277
[LinkedIn](#) (13K+ followers) • [GitHub](#) • [Medium](#)
goortani.com • visionzlab.com

Seeking Staff/Principal IC Role at FAANG or FAANG-Tier Company • Cutting-Edge AI/ML Systems

Professional Summary

Staff-level AI/ML engineer with **25+ years** building production systems at scale. Currently **Solution Architect at Uber**, designed and built **ELLE AI Decision Engine** - production LLM system processing **millions of decisions** across Uber's global platform (won multiple innovation awards). Deep expertise in **distributed systems** (Go microservices, Cadence workflows, Kafka), **production LLM applications** (LangChain, RAG, AI Agents), and **full-stack engineering** (React, TypeScript, Python, cloud-native). Proven ability to architect and ship complex systems serving millions of users with sub-200ms latency and 99.95%+ uptime.

Seeking **Staff/Principal IC role** at FAANG or FAANG-tier company working on cutting-edge AI/ML systems.

Tools, Frameworks, Libraries, Languages, Databases

AI/ML & LLM Frameworks

LangChain • LangGraph • FastMCP • MCP Development • CrewAI • Ollama • Prompt Engineering • Agentic Workflows • AI Agents • RAG • Multi-Agent Systems • ReAct Patterns • Self-Reflection • Model Fine-Tuning

Agentic Coding & AI Tools

Claude Code • Cursor • GitHub Copilot • Codex • Aider • AI-Assisted Development • MCP Server Development • 3-5x Productivity Gains

LLM Platforms

OpenAI (GPT-4, GPT-3.5) • Anthropic Claude (Sonnet, Opus) • Google Gemini • Cohere • Meta LLMs • Hugging Face Models • Open Source LLMs • Model Evaluation • Model Selection

Vector Databases & Embeddings

OpenSearch • Pinecone • Qdrant • Chroma • Weaviate • Embedding Models • Semantic Search • Vector Similarity Search • Re-ranking

Backend Frameworks

Python (FastAPI, LangFx, Pydantic, Uvicorn, async/await) • Go (YARPC, gRPC, microservices) • Java (Spring Boot, Spring Cloud) • Node.js (Express, NestJS) • C# (.NET Core)

Distributed Systems & Workflow

YARPC (Uber RPC) • gRPC • Apache Kafka • Cadence Workflows • Temporal • Apache Thrift • Protocol Buffers • Service Mesh (Istio) • Circuit Breakers • Distributed Tracing • Event-Driven Architecture

Frontend Frameworks

React 18 • TypeScript 5.7 • Fusion.js (Universal Rendering) • NextJS • TanStack React Query • Redux • Angular • BaseUI • Styletron • Server-Side Rendering • Micro-Frontends

Cloud Platforms

Google Cloud (GCP Cloud Run, Firestore, Firebase, GKE) • AWS (EC2, Lambda, S3, RDS, DynamoDB, ECS, EKS) • Azure (VMs, Functions, CosmosDB, AKS) • Serverless Architecture

Container & Orchestration

Docker • Kubernetes (K8S) • Helm • Istio • Rancher • Google Kubernetes Engine • Container Orchestration • Service Discovery • Health Checks

API & Integration

GraphQL • gRPC • REST • Protocol Buffers • Apache Thrift • Swagger/OpenAPI • OAuth2/OIDC • JWT • Webhooks • API Gateway • Rate Limiting

Data & Messaging

Apache Kafka (Kafka Streams, Kafka Connect) • Cadence • Temporal • Apache Spark (PySpark) • Presto • Hive • HDFS • Airflow • Stream Processing • Real-Time Analytics

Databases

Cassandra (Large-Scale Distributed) • MySQL • PostgreSQL • MongoDB • Redis (Caching, Rate Limiting) • DynamoDB • Elasticsearch • Firestore • CouchDB • SQL Server

DevOps & CI/CD

Bazel (Large-Scale Monorepo) • GitHub Actions • Jenkins • Bamboo • Circle CI • BuildKite • Terraform • Ansible • Infrastructure as Code • GitOps • Continuous Deployment

Monitoring & Observability

M3 (Uber Metrics Platform) • Prometheus • Grafana • Datadog • Splunk • Zap Logging • Tally Metrics • Jaeger (Distributed Tracing) • Kibana • Application Performance Monitoring

Security & Identity

Charter (Uber IAM) • SPIFFE/SPIRE (Workload Identity) • RBAC • OAuth2/OIDC • JWT • mTLS • PKI • Encryption • Security Middleware

Build & Tooling

Bazel • Yarn Workspaces • Vite • Webpack • Monorepo Architecture • Code Generation • Static Analysis

Skills and Achievements

Distributed Systems & Scale:

- Production Systems at Scale: Built systems serving millions of users with sub-200ms p95 latency and 99.95%+ uptime
- Microservices Architecture: Designed and deployed 40+ Go microservices using YARPC, gRPC, and service mesh patterns
- Event-Driven Architecture: Apache Kafka for real-time streaming, Cadence/Temporal for workflow orchestration
- Distributed Tracing: Jaeger integration for debugging complex multi-service requests across 30+ microservices
- Performance Optimization: Achieved 3x latency reduction on LLM inference, 60% API response time reduction with caching
- Reliability Engineering: Circuit breakers, health checks, fallback strategies for resilient service communication
- Data Pipelines: Apache Spark processing 2M+ events/day, Kafka streams for real-time analytics

AI/ML & Production LLM Systems:

- Production LLM Systems: Multi-agent architectures, RAG pipelines, semantic search, prompt engineering at Uber scale
- AI Agents: ReAct patterns, self-reflection, multi-agent orchestration with LangGraph (6+ specialized agents)
- RAG Systems: OpenSearch vector store indexing 1M+ documents, semantic search with re-ranking
- Model Integration: OpenAI, Anthropic Claude, Google Gemini with fallback strategies and circuit breakers
- Prompt Engineering: 92% accuracy on privacy classification tasks with optimized prompts and few-shot learning
- Vector Databases: OpenSearch, Pinecone, Qdrant for semantic search and document retrieval
- ML Pipelines: Data preprocessing, feature engineering, model evaluation, continuous monitoring

Technical Leadership & Impact:

- Staff-Level IC: Technical leadership without direct reports, cross-team collaboration with 5+ platform teams
- System Design: Architected ELLE AI Decision Engine processing millions of decisions with 70% efficiency gain
- Innovation Awards: Won multiple Uber internal awards for production AI systems and automation workflows
- Quality Excellence: 96% test coverage across codebases, zero security incidents in 18+ months
- Cross-Functional Collaboration: Partnered with Identity, Workflows, Data, Messaging, Observability teams
- Technical Writing: Designed comprehensive technical documentation, architectural decision records (ADRs)
- Code Reviews: Conducted 500+ code reviews ensuring quality, security, and architectural consistency

Full-Stack Engineering:

- Frontend Excellence: React 18, TypeScript, universal rendering (Fusion.js), SSR achieving 40% faster page loads
- Backend Development: Python (FastAPI, async/await), Go (microservices), Java (Spring Boot)
- Type Safety: Protocol Buffers for RPC, TypeScript for frontend, Pydantic for Python
- Testing: 96% code coverage with Jest, Pytest, React Testing Library, Playwright E2E tests
- Build Systems: Bazel monorepo builds, Yarn Workspaces, CI/CD with GitHub Actions
- Observability: M3 metrics, Zap logging, Jaeger tracing integrated across all applications

Professional Experience

Uber I Solution Architect

ELLE (EngsecLLM Engine) - AI Decision Engine - Staff-Level Impact

Scale & Complexity:

- Built production LLM system processing **millions of security/privacy decisions** across Uber's global platform
- Architected system handling **100K+ requests/month** with **sub-200ms p95 latency**
- Designed **multi-agent architecture** with 6+ specialized agents (Technical Privacy Review, Security Guidance, Legal Review, DSAR Automation, Data Classification, Compliance Checking)
- Implemented **RAG system** with OpenSearch vector store indexing **1M+ documents** (security policies, privacy regulations, compliance frameworks)

Technical Leadership:

- **System Design:** Designed distributed architecture using Python microservices, Go authorization layer, Cadence workflows, Kafka event streaming
- **Cross-Team Collaboration:** Partnered with 5+ platform teams (Identity, Workflows, Data, Messaging, Observability) to integrate with Uber infrastructure
- **Performance Optimization:** Optimized LLM inference achieving 92% accuracy on privacy classification with 3x latency reduction
- **Observability:** Built comprehensive monitoring with Zap logging, Tally metrics, M3 dashboards, Jaeger distributed tracing

Technical Achievements:

- **Won multiple internal innovation awards** for automating security/privacy workflows (70% reduction in manual review time)
- **96% test coverage** across authorization middleware, LLM workflows, and RAG pipelines
- **Zero security incidents** since launch (18+ months in production)
- **99.95% uptime** serving global compliance teams (500+ users)

Technologies: Python (FastAPI, LangChain, LangGraph), Go (YARPC, gRPC), OpenAI API, OpenSearch (vector store), Cadence workflows, Kafka, Cassandra, Redis, Protocol Buffers, Bazel, Kubernetes, M3, Jaeger

System Design Patterns:

- Multi-agent orchestration with LangGraph
- RAG with semantic search and re-ranking
- Circuit breakers and fallback strategies for LLM calls
- Authorization middleware with dual inbound/outbound protection
- Event-driven architecture with Kafka
- Distributed tracing for debugging microservices

Distributed Systems & Platform Engineering - Scale & Reliability

Contributed to 40+ Engineering Security projects in Uber's Go monorepo, focusing on distributed systems, authorization, and platform services:

Major Platform Contributions:

1. Charter (Authorization Platform):

- Integrated Charter Decision Service (Uber's IAM equivalent) for fine-grained access control
- Designed RBAC policies protecting 100+ GenAI platform APIs
- Built authorization middleware with **96% test coverage**
- Implemented dual middleware architecture (inbound/outbound) achieving comprehensive endpoint coverage

2. SPIRE/SPIFFE (Workload Identity):

- Implemented service-to-service authentication using SPIFFE/SPIRE
- Integrated mTLS for secure communication between microservices
- Designed workload identity attestation for containerized services

- Built PKI infrastructure for certificate rotation and management

3. USSO (Uber SSO):

- Integrated Single Sign-On across all internal tools
- Implemented OAuth2/OIDC flows with JWT validation
- Built session management and token refresh logic
- Designed secure token storage and rotation mechanisms

4. Distributed Systems Architecture:

- Go microservices using YARPC (Uber's RPC framework) and gRPC
- Cadence workflows for long-running asynchronous processes
- Kafka messaging for event-driven communication
- Cassandra and MySQL for distributed data storage
- Redis for caching and rate limiting

Technical Achievements:

- Solved complex **circular dependency challenges** using MCP Gateway pattern with fx.Lifecycle hooks
- Implemented **dual middleware architecture** (inbound/outbound) achieving comprehensive endpoint coverage
- Built **distributed tracing integration** with Jaeger for debugging multi-service requests spanning 30+ services
- Designed **health check and circuit breaker** patterns for resilient service communication
- Achieved **99.95% uptime** for authorization services serving 10K+ requests/second

Technologies: Go (Golang), YARPC, gRPC, Protocol Buffers, Apache Thrift, Charter, SPIFFE/SPIRE, authfx, OAuth2, JWT, Cadence, Kafka, Cassandra, MySQL, Redis, Bazel, Docker, Kubernetes, Zap, Tally, Jaeger

Full-Stack Engineering at Scale - 9 Enterprise Web Applications

Led full-stack development of 9+ mission-critical privacy web applications serving Uber's global compliance teams:

Scale & Performance:

- **Fusion.js universal rendering** achieving **40% faster page load times** (SSR + code splitting)
- **Type-safe RPC communication** via Protocol Buffers for 30+ backend services
- **Comprehensive testing:** 34+ Jest test suites with **96% code coverage**, Playwright E2E tests
- **Global deployment:** Serving users across 70+ countries with sub-second page loads
- **Micro-frontends:** Modular deployment enabling independent team velocity

Applications:

1. **Elle Web** - EngsecLLM Engine Frontend (React 18, TypeScript, LangChain UI integration)
2. **IRIS Web** - Privacy Web Frontend (GDPR/CCPA compliance automation)
3. **DSAR Web** - Subject Access Request Automation (60% efficiency gain)
4. **UDE (User Data Explorer)** - Data discovery and lineage tracking
5. **Consents Manager Web** - Contextual consent management (GDPR/CCPA compliant)
6. **Web CMP** - IAB TCF v2 Compliant Consent Management Platform (70+ countries)
7. **TPRM QC Copilot Web** - Third-Party Risk Management quality control
8. **Agentic AI Observability Web** - Real-time AI agent monitoring and debugging
9. **Design Review Web** - Privacy design review workflows

Technical Stack:

- React 18, TypeScript 5.7, Fusion.js (universal rendering)
- TanStack React Query (data fetching), Redux (state management)
- BaseUI design system, Styletron CSS-in-JS
- Protocol Buffers (RPC), GraphQL (Apollo Client)
- Jest (unit tests), Playwright (E2E), React Testing Library
- Bazel (monorepo builds), Yarn Workspaces, Vite

Architecture Patterns:

- Universal rendering with SSR and code splitting
- Micro-frontends for modular deployment
- Type-safe API contracts via Protocol Buffers
- Feature flags with Flipr for gradual rollouts
- Observability with M3 metrics and Zap logging

Python Monorepo - 50+ Security Infrastructure Projects - AI/ML & Data Engineering

Led development of 50+ security infrastructure projects spanning AI/ML, data engineering, and automation:

AI/ML Projects:

- LLM-powered security automation using LangChain, LangGraph, FastMCP (Model Context Protocol)
- RAG systems for security policy search (OpenSearch vector store, semantic embeddings)
- Multi-agent systems with self-reflection and ReAct patterns
- Container image remediation assistant (LangFx service with JIRA integration)

Data Engineering:

- Apache Spark pipelines for security event processing (2M+ events/day)
- Kafka streaming for real-time security alerts and anomaly detection
- Presto SQL queries for compliance reporting and analytics

Technologies: Python 3.9/3.11, LangChain, LangGraph, FastMCP, Apache Spark, Kafka, Presto, OpenSearch, Pinecone, Qdrant, Bazel, pytest

Home Depot Canada | Lead Solution Architect

Dec 2017 - Dec 2019 | Toronto, Canada

Scale & Impact:

- **Team Leadership:** Led team of **10+ onshore and 50+ offshore developers** for homedepot.ca website
- **Customer 360 Platform:** Unified customer data platform serving **15M+ customers** with personalized experiences
- **Revenue Impact:** Enabled **\$50M+ in targeted marketing** revenue through personalized customer insights
- **Performance:** Improved customer satisfaction by 25% through personalized recommendations

Technical Achievements:

- Designed **microservices architecture** (25+ services) for real-time customer data processing
- Built **Kafka/Spark pipeline** processing **2M+ events/day** for real-time analytics
- Implemented **Angular Universal Server-Side Rendering** for SEO and social sharing (40% traffic increase)
- Architected **caching layer** leveraging Redis, reducing API response times by 60%
- Led **mobile DevOps automation** using Jenkins, Fastlane, achieving 90% deployment automation

Technologies: Java Spring Boot, Kafka, Cassandra, Redis, React, Angular, AWS, Kubernetes, Jenkins

Canada Life | Enterprise Architect

Jan 2019 - Jan 2021 | Toronto, Canada

Scale & Impact:

- **API Platform:** Designed RESTful API platform serving **100+ microservices** across insurance operations
- **Mobile Architecture:** Led iOS/Android native development for customer-facing mobile apps (2M+ users)
- **Microservices Migration:** Architected migration from monolith to microservices (Kubernetes, Docker, Rancher)

Technical Achievements:

- Designed **event-driven architecture** using Kafka for real-time policy updates and claims processing
- Built **API gateway** with rate limiting, authentication, and circuit breakers
- Implemented **Kubernetes cluster** with automated scaling and self-healing

Technologies: Java Spring, Kubernetes, Docker, Rancher, Kafka, PostgreSQL, Redis, iOS (Swift), Android (Kotlin)

TD Bank | Senior Manager, Enterprise Applications

Oct 2014 - Jun 2015 | Toronto, Canada

Scale & Impact:

- **Team Leadership:** Managed team of **12+ developers** delivering BI projects for banking operations
- **Payment Processing:** Real-time payments platform processing **\$2B+/year** with 99.99% uptime
- **Omnichannel Banking:** React Native mobile + React web serving **8M+ users**

Technical Achievements:

- Designed **Banking Services Data Mart** for OMEGA-PEGA project (PowerDesigner, SSIS, SSRS, SSAS)
- Implemented **microservices migration** (25+ services) reducing deployment time from weeks to hours
- Built **real-time fraud detection** system using streaming analytics

Technologies: Java Spring, React, React Native, PostgreSQL, Redis, Docker, Jenkins, SQL Server, SSIS/SSAS/SSRS

CAA South Central Ontario I Solution Architect

Apr 2012 - Nov 2013 | Thornhill, Canada

Enterprise Architecture:

- Designed **complex Data Warehouse projects** in Finance, Insurance, and Travel industries
- Implemented **ESB web projects** using SOA and TOGAF patterns (WCF, Web Services, .NET 4.5)
- Led **SQL Server Infrastructure** design and upgrades (clustering, performance optimization, HA/DR)
- Architected **Enterprise Data Dictionary** solution for data warehouse governance

Technologies: SQL Server 2012, SSIS/SSAS/SSRS, QlikView, C#, .NET 4.5, WCF, TOGAF, SharePoint, Windows Azure

Open Source & Community

Contributions:

- LangChain ecosystem contributions (bug fixes, feature development, documentation)
- Active in AI Engineer community (AI Engineer Summit, LangChain Discord, local meetups)
- Open source tools for AI/ML workflows and distributed systems patterns

Thought Leadership:

- **LinkedIn:** 13K+ followers, AI/ML thought leader focusing on production LLM systems, distributed systems
- **Medium:** [@FrankGoortani](#) - Published articles on LangChain, RAG systems, distributed architectures
- **Speaking:** Available for conference talks on production AI systems, distributed systems, and scale

Education and Professional Development

M.Sc. in Management

AmirKabir University of Technology (Tehran Polytechnic), Iran | 2003

B.Sc. in Computer Software Engineering

AmirKabir University of Technology (Tehran Polytechnic), Iran | 2001

Certifications:

- Certified Data Scientist, Johns Hopkins University, 2015
- PMP Certified (Project Management Professional), PMI, 2013
- TOGAF 9.1 Certified, The Open Group Architecture Framework, 2013
- ITIL Foundation, Mind Leaders, 2013
- MCPD - Web Application Development 2008, Microsoft, 2012
- MCITP - Database Developer 2008, Microsoft, 2012
- Agile Project Management Strategy and Analysis, 2013
- Data Warehousing Management, Mind Leaders, 2013
- OWASP Top 10, Security Compass, 2012

Key Strengths

- **Staff-Level IC:** Technical leadership without direct reports, cross-team collaboration, architectural decision-making

- **Scale & Performance:** Systems serving millions of users with sub-200ms latency, 99.95%+ uptime
- **Distributed Systems:** Deep expertise in Go microservices, Kafka, Cadence, distributed tracing, service mesh
- **AI/ML Production:** Shipped production LLM systems (LangChain, RAG, AI Agents) at Uber scale
- **Full-Stack Depth:** React/TypeScript frontend + Python/Go backend + cloud infrastructure
- **Quality-Focused:** 96% test coverage, zero security incidents, comprehensive observability
- **System Design:** Architected complex distributed systems, API platforms, workflow orchestration